# Appendix F: Simulation of Sister Groups and Outgroups

Chapter 10 features a comparison of (subsets of) the Funnel Beaker West and Corded Ware body of knowledge with two simulated sister groups (sim_Vl and sim_FBN) and one simulated outgroup (sim_Odisha, see Tab. 10.2). This appendix discusses the simulation procedure behind these simulated groups and the choices made in the simulation process. The selection of these datasets is discussed in Section 10.2.

## 1 Setting up the Simulation

The algorithm for guided random path generation is introduced in Section 4.2 (see tab. 4.2). This algorithm can (i.a.) generate paths based on quantitative or qualitative information about ceramic production processes in other studies. This procedure requires three preparatory steps, which partially depend on the base dataset about ceramic production.

The first step is translation. The use of datasets from other studies in guided random simulation requires them to be compatible with the network in Figure 4.1. Therefore, the base data has to be formatted following the vocabulary from Roux (2019). This process is usually straightforward, and involves a re-interpretation of the descriptions and documentation of techniques (if present) in the base dataset. The paragraphs below discuss these interpretations for each of the base datasets.

The second step consists of plugging this information into the total *chaîne opératoire* (see Fig. 4.1). The complexity of this step depends on the data. Two factors are crucial in the set-up the simulation: the degree of fragmentation, and the availability of quantitative information about the use of techniques.

The degree of fragmentation refers to the availability of complete specific *chaînes opératoires*. We can distinguish three states of fragmentation. A dataset has low fragmentation if it consists of complete specific *chaînes opératoires* (see Ch. 6, 7). A dataset with medium fragmentation contains path segments, but not complete specific *chaînes opératoires*. A path segment is a fragment of a complete specific *chaîne opératoire*. For example, the combination of techniques applied from finishing stage to surface treatment is a path segment. Lastly, a dataset has high fragmentation if it contains a list of the techniques used during the production process, but no indication of

their ordering. Keep in mind that fragmentation may not be uniform in a dataset: a study may report both path segments and lists of techniques (e.g. Behura 1978).

The degree of fragmentation determines the amount of effort required to link up these fragments into complete specific *chaînes opératoires*. Datasets with low fragmentation require no further effort, but the links between techniques or path segments need to be inferred in a dataset with medium or high fragmentation. The procedure applied below joins these fragments by fully connecting the observed segments and techniques. This means allowing for all possible links between the techniques and/or segments. For example, if two finishing techniques and two decorative techniques occur on wet clay according to a study, then all possible links within and between these techniques in the finishing and decorative stage are allowed for in the network. If undecorated vessels also occur, we furthermore allow for all possible links between the finishing techniques and (f.e.) drying to leather-hard. A more parsimonious approach to joining fragments and techniques is also possible, for example using ant colony optimisation (cf. Dorigo and Stützle 2004; Mirjalili 2019). However, the advantage of fully connected path segments over optimisation is that we do not impose restraints on past potters. By fully connecting the path segments we assume potters could perform any combination of the (series of) techniques they were familiar with, whereas optimisation would only allow for certain optimal combinations. We return to some of the implications of fully connected segments below.

The second factor in the set-up of the simulation is the availability of quantitative data on ceramic production processes. If quantitative information is present, we know something about the preferred (combinations of) techniques. This information can be incorporated in the network from Figure 4.1 by adjusting the edge weights, similar to the procedure for complete specific *chaînes opératoires* (see Section 4.2). However, a different approach is necessary if there is no or partial quantitative information on production techniques. For example, a study may provide tallies of sherds with particular decorative techniques and a narrative account of roughing-out techniques. In these cases, it is preferable to set the value of all links which occur or have been inferred (see above) to 1 and the values of all other links to 0. This binary value of the links ensures the algorithm for guided path generation can run without

incurring a bias due to (f.e.) overrepresentation of decorative techniques and surface treatment (see below).

The result of the above two steps is an event space (see Section 4.2). This event space captures all the possible combinations of techniques potters would be familiar with according to the base dataset. If quantitative information was available, the event space also contains information about the combinations of techniques potters preferred. If quantitative information was absent or partial, the event space only shows the possible sequences. This event space is referred to here as a random seed and stands at the basis of guided random path generation.

The final step in the simulation procedure consists of applying the algorithm for guided random path generation to this random seed. Recall that this algorithm looks both at the syntax (see Ch. 4) and at the edge weight (see Tab. 4.2) when selecting a neighbouring node as the next node during the generation of a path. The chance to select a specific neighbouring node is proportional to the edge weight, i.e. number of paths in the dataset which feature that particular combination of techniques (see Section 4.2). As such, the algorithm reproduces the choices of past potters in the generated paths. At this point, the advantages of a dataset with quantitative information and low fragmentation are apparent: this ensures the selection of a neighbouring node is always guided by information about the choices of past potters. As a result, the generated paths resemble the choices of these potters more closely. However, if the fragmentation is high and quantitative information is absent in a dataset, then the binary edge weight and fully connected path segments enable us to generate paths with the fewest possible additional assumptions about the choices of potters. At any point, the algorithm considers only the techniques potters were familiar with and has an equal chance to select any of these techniques.

The generated paths are not a perfect representation of the production processes within the base dataset. Consider a body of knowledge with specific *chaînes opératoires* which are all identical, except one half features clay coating between drying to leather hard and drying to dry consistency, as well as painting after firing. The other half of the specific *chaînes opératoires* does not feature these techniques. Guided random path generation over such a body of knowledge does not necessarily replicate this connection between clay coating and painting, but generates all possible variants of

these specific *chaînes opératoires* (e.g. no clay coating, but painted, only clay coated, etc.). Therefore, the specific *chaînes opératoires* which result from guided random path generation are best thought of as representing all possible variation within the confines of the original dataset (see Section 3.2) and the syntax (see Section 4.1), but not as actual specific *chaînes opératoires* one might find from these potters. The prefix 'sim_' is used to indicate this status of the generated paths in Chapter 10.

All paths in the simulated outgroups and sister groups were generated with a default value of 0 for undetected links, a multiplier of 1 for detected links, a recursion factor of 0, and 1,000 iterations. These settings ensure the generated paths remain representative of the combinations of techniques observed in actual data (see Code).

The above three steps (translation, setting up the event space, and applying the algorithm) were applied to all three base datasets for the sister groups and outgroups. The sections below discuss each base dataset, and the conversion into a random seed.

## 3 Random Seed for Sim_Vl

The first simulated body of knowledge in Chapter 10 is based on a study of ceramic production processes at the site Vlaardingen-Arij Koplaan by Stet (2021; Van Beek 1990). The ceramic vessels are associated with the Vlaardingen group (see Section 2.1; 10.2). The dataset from Stet (2021) has three key advantages over other studies of Vlaardingen ceramic technology (i.e. Beckerman 2015; Kroon *et al.* 2019): It follows the study protocol from Roux (2019), presents photographic and textual evidence for techniques (see source criticism in Sections 6.1; 7.1), and provides quantitative information about Vlaardingen ceramic technology with medium fragmentation. The downside of this dataset is that the material is relatively young (ca. 2671-2129 cal BCE; cf. Stet 2021 pp. 24–5) and stems from a single site. This means the dataset may not be representative of the variation in Vlaardingen ceramic technology as a whole. However, a comparison with broader studies from Beckerman (2015), and Kroon *et al*. (2019) shows the results are at least typical for Vlaardingen ceramic technology. Therefore, the base dataset is best seen as indicator for Vlaardingen ceramic technology in the third millennium BCE.

The translation of the base dataset is relatively straightforward given the quality of the documentation and use of the study protocol from Roux (2019). In general, the

roughing-out techniques for the base consist of modelling a clay mass for the centre of the base and supplementing this mass with adjacent coils. This procedure has left bases with an irregular topography and a circular fissure visible on the base exterior (cf. Stet 2021 p. 36, Fig. 6). The breakage pattern of the vessel wall and break profiles indicate the use of the coiling technique for roughing out these parts of the vessel (ibid. pp. 33-4, Figs. 4-5). The production technique during the preforming stage consists of continuous pressure applied by hand to wet clay. The use of this technique is evidenced by the appearance of elongated finger-sized hollows in vertical and diagonal orientation on various vessel parts (ibid. pp. 37-8, Fig. 7). In addition, there are handles which lack traces of plugs and have been applied to the vessel wall while wet (ibid. p. 38, Figs. 8-9).

Stet (2021) reports three techniques during the finishing stage: smoothing wet clay with a hard tool, smoothing wet clay with a soft tool, and smoothing leather hard clay. However, the photographic documentation of smoothing wet clay with a hard tool (ibid. Fig. 10) and smoothing on leather hard clay (ibid. Appendix 10) show traces of a different operation. These photos show parallel horizontal striations with scalloped edges and surfaces with compact microtopography, inserted particles, and glossy aspect. Such traces are typical for burnishing (cf. Roux 2019 p. 201). The typical traces of wet smoothing (i.e. irregular microtopographies, matt aspect, protruding particles, and some threaded striation Roux 2019 p. 196) do occur (ibid. Fig. 11), as well as traces of the use of hard tools (ibid. Fig. 10, 12) in the shape of striations with over-thicknesses. As such, the finishing techniques point towards wet smoothing with hard and soft tools.

The only decorative techniques found on the Vlaardingen vessels are excisions in the vessel wall below the rim. The tools used to make these excisions had a round cross-section, and left thickened edges. Traces of surface treatment cut across the edges of the excisions (cf. Stet 2021 pp. 45–6, Fig. 14). As such, these excisions were made on wet clay, prior to surface treatment.

The traces of surface treatment all point towards the use of burnishing on soft leather hard clay (see above, cf. Stet 2021 p. 43, Fig. 13). As such, we can also infer the presence of a drying stage prior to surface treatment. No traces of other operations prior to firing were detected, meaning drying to dry consistency occurred after burnishing. The firing process itself most often resulted in oxidation of the exterior margin, with a sharp transition to a reducing core and interior margin. Vessels with

oxidised interior margins also appear (cf. Stet 2021 p. 49, Tab. 11, Fig. 15). As such, the vessels were most likely subjected to open firing. Stet (2021) documents no traces of post-firing operations.

The dataset from Stet (2021) has medium to high fragmentation: the operations are either reported stage-by-stage or as paths segments (ibid. p. 51-2, Tab. 9). Furthermore, there is quantitative information on the number of sherds with specific combinations of techniques (ibid. Tabs. 4-11) and the minimum number of individuals in the ceramic assemblage (n=71, Stet 2021 p. 33). As such, we can calculate the edge weights in the network by assuming the sherds in the assemblages are equally distributed over these 71 vessels (see Tab. 1). Figure 1 presents the resulting event

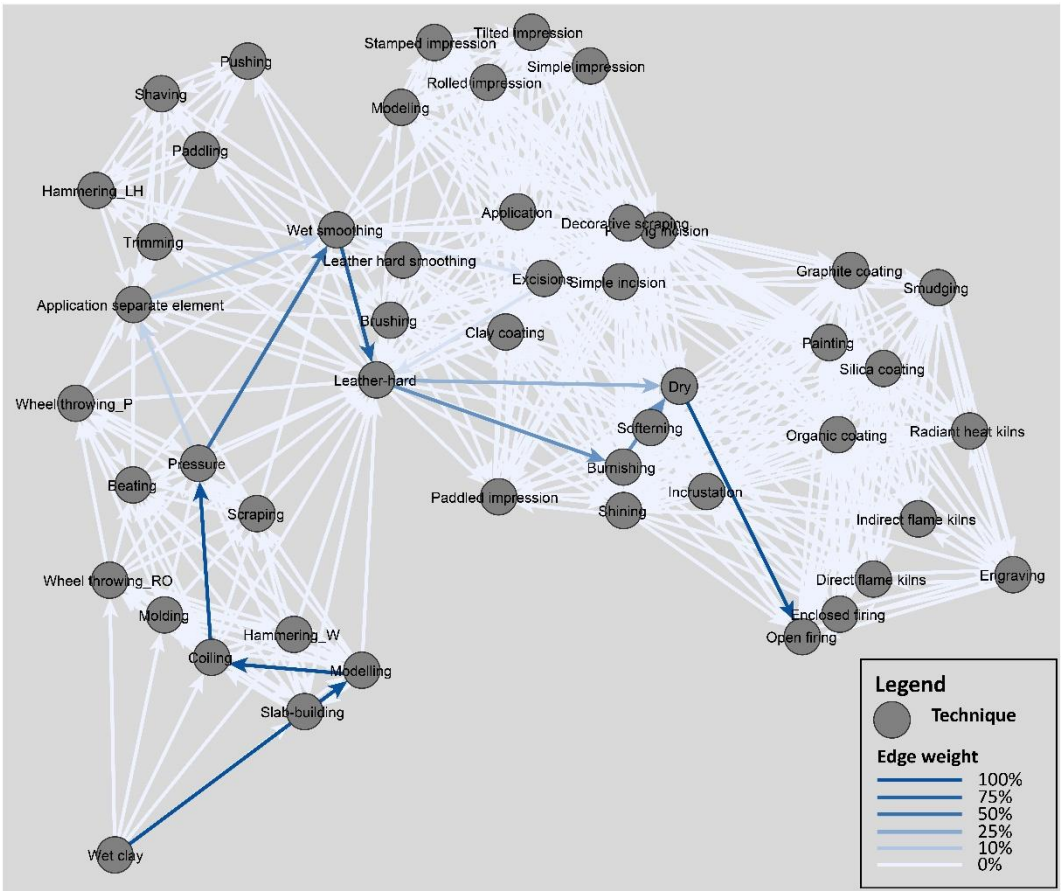



Figure 1: Random seed for sim_Vl. The blue edges represent the sequences of techniques observed in the Vlaardingen ceramics from the site Vlaardingen-Arij Koplaan by Stet (2021). The darker the colour of the edges, the higher the percentage of vessels which exhibit this combination of techniques.

Table 1: Edge lists for the random seeds of sim_Vl, sim_FBN, and sim_Odisha. Only the edges which occur at least once in these three datasets are shown. The value of the edges stems from the base dataset.

| **Edge** | | **Random seed for:** | | |
|---|---|---|---|---|
| Source | Target | sim_Vl | sim_FBN | sim_Odisha |
| *Node* | *Node* | *%* | *binary* | *binary* |
| Wet clay | Modelling | 100 | 1 | 1 |
| Wet clay | Molding | 0 | 0 | 1 |
| Wet clay | Wheel throwing_RO | 0 | 0 | 1 |
| Coiling | Leather-hard | 0 | 0 | 1 |
| Coiling | Scraping | 0 | 0 | 1 |
| Coiling | Pressure | 100 | 1 | 1 |
| Modelling | Coiling | 100 | 1 | 1 |
| Modelling | Pressure | 0 | 0 | 1 |
| Modelling | Beating | 0 | 0 | 1 |
| Molding | Coiling | 0 | 0 | 1 |
| Molding | Leather-hard | 0 | 0 | 1 |
| Wheel throwing_RO | Wheel throwing_P | 0 | 0 | 1 |
| Leather-hard | Shaving | 0 | 0 | 1 |
| Leather-hard | Paddling | 0 | 0 | 1 |
| Leather-hard | Dry | 44 | 1 | 0 |
| Leather-hard | Burnishing | 56 | 1 | 0 |
| Leather-hard | Simple incisions | 0 | 1 | 0 |
| Scraping | Leather-hard | 0 | 0 | 1 |
| Scraping | Application separate element | 0 | 1 | 0 |
| Scraping | Wet smoothing | 0 | 1 | 0 |
| Pressure | Leather-hard | 0 | 0 | 1 |
| Pressure | Scraping | 0 | 1 | 0 |
| Pressure | Application separate element | 13 | 1 | 0 |
| Pressure | Wet smoothing | 87 | 1 | 1 |
| Beating | Wet smoothing | 0 | 0 | 1 |
| Shaving | Paddling | 0 | 0 | 1 |
| Shaving | Application separate element | 0 | 0 | 1 |
| Shaving | Leather-hard smoothing | 0 | 0 | 1 |
| Paddling | Application separate element | 0 | 0 | 1 |
| Paddling | Leather-hard smoothing | 0 | 0 | 1 |
| Wheel throwing_P | Wet smoothing | 0 | 0 | 1 |
| Application separate element | Wet smoothing | 13 | 1 | 0 |

| | | | | |
|---|---|---|---|---|
| Application separate element | Leather-hard smoothing | 0 | 0 | 1 |
| Wet smoothing | Leather-hard | 92 | 1 | 1 |
| Wet smoothing | Simple impression | 0 | 1 | 0 |
| Wet smoothing | Simple incision | 0 | 1 | 0 |
| Wet smoothing | Excisions | 8 | 1 | 0 |
| Wet smoothing | Application | 0 | 1 | 0 |
| Leather-hard smoothing | Clay coating | 0 | 0 | 1 |
| Leather-hard smoothing | Paddled impression | 0 | 0 | 1 |
| Leather-hard smoothing | Simple incision | 0 | 0 | 1 |
| Leather-hard smoothing | Excisions | 0 | 0 | 1 |
| Leather-hard smoothing | Application | 0 | 0 | 1 |
| Dry | Open firing | 100 | 1 | 1 |
| Dry | Enclosed firing | 0 | 0 | 1 |
| Dry | Indirect flame kiln | 0 | 0 | 1 |
| Burnishing | Leather hard smoothing | 56 | 1 | 1 |
| Shining | Leather hard smoothing | 0 | 0 | 1 |
| Clay coating | Burnishing | 0 | 0 | 1 |
| Clay coating | Shining | 0 | 0 | 1 |
| Smudging | Incrustation | 0 | 1 | 0 |
| Simple impressions | Leather-hard | 0 | 1 | 0 |
| Simple impressions | Simple incision | 0 | 1 | 0 |
| Paddled impression | Clay coating | 0 | 0 | 1 |
| Simple incision | Leather-hard | 0 | 1 | 0 |
| Simple incision | Burnishing | 0 | 1 | 0 |
| Simple incision | Clay coating | 0 | 0 | 1 |
| Simple incision | Simple impression | 0 | 1 | 0 |
| Excisions | Leather-hard | 8 | 1 | 0 |
| Excisions | Simple impression | 0 | 1 | 0 |
| Excisions | Incrustation | 0 | 0 | 1 |
| Incrustation | Burnishing | 0 | 0 | 1 |
| Application | Leather-hard | 0 | 1 | 0 |
| Application | Clay coating | 0 | 0 | 1 |
| Application | Simple impression | 0 | 1 | 0 |
| Application | Simple incision | 0 | 1 | 0 |
| Open firing | Organic coating | 0 | 0 | 1 |
| Open firing | Silica coating | 0 | 0 | 1 |
| Open firing | Smudging | 0 | 1 | 0 |
| Open firing | Painting | 0 | 0 | 1 |

| | | | | |
|---|---|---|---|---|
| Open firing | Incrustation | 0 | 1 | 0 |
| Enclosed firing | Organic coating | 0 | 0 | 1 |
| Enclosed firing | Silica coating | 0 | 0 | 1 |
| Enclosed firing | Painting | 0 | 0 | 1 |
| Indirect flame kiln | Organic coating | 0 | 0 | 1 |
| Indirect flame kiln | Silica coating | 0 | 0 | 1 |
| Indirect flame kiln | Painting | 0 | 0 | 1 |

space for Vlaardingen ceramic technology. This is the random seed for the generation of sim_Vl (see Ch. 10).

## 3 Random Seed for Sim_FBN

The second group of simulated specific *chaînes opératoires* is sim_FBN (see Section 10.2). The random seed for this group based on Koch's (1998) study of ceramics from Funnel Beaker North bog deposits on the islands Zealand, Møn, Lolland, and Falster (Denmark). These depositions date between 3950 and 2900 BCE (Koch 1998 p. 15).

Koch (1998 p. 192) argues the vessels in these bog deposits are a selection from the broader range of vessel types found on other Funnel Beaker North sites (see Fig. 2.7). Specifically, beakers would be overrepresented in bog depositions relative to other site types. Unfortunately, no additional data exists to assess whether this overrepresentation of a specific vessel type also leads to overrepresentation of a specific production technique or method. However, the various Funnel Beaker West vessel types do not differ significantly in terms of production processes (see Ch. 6) and Koch (1998 p. 125) suggests the same applies Funnel Beaker North. Therefore, the base dataset is treated as representative for Funnel Beaker North here, but additional studies of Funnel Beaker North ceramic technology would be preferable to confirm this.

The problems with the technical observations from Koch (1998) have been discussed in Section 6.1. The focus here is on translating the observations into the vocabulary from Roux (2019). Koch (1998 p. 125; 1987 pp. 111–2) argues based on observations of the microstructure and voids in cross-sections that the bases of Funnel Beaker North vessels consist of modelled clay disks. The breakage pattern, break profiles, and microstructure of vessel walls suggest the coiling techniques was used

during the roughing-out stage with different joining methods appearing in the same vessels (ibid. Fig. 92, 96).

Koch (1998 p. 125) mentions the use of elongated cylinders to preform vessels with percussion. However, these elongated depressions in vertical orientation on the vessel wall are typical for preforming by continuous pressure with hands, rather than percussion (cf. Roux 2019 pp. 174–5, 176). Further striations in horizontal orientation at the shoulder of vessels probably indicate the use of scraping (cf. Koch 1998 p. 127, Fig. 97) during preforming. In addition, Koch (1998 Pls. 30, 46, 90, 97) depicts several handles which have broken off from the vessel wall. Some of these handles exhibit plugs which were used to affix them to the vessel wall, others lack such traces and were likely applied by pressure (see also Ch. 6).

The evidence for techniques used during the finishing stage is sparse. Koch's (1998 p. 127) descriptions suggest some vessel surfaces exhibit more irregular microtopographies and protruding particles, which were either partially or fully covered. These traces are typical for wet smoothing. However, there is no further evidence for operation during the finishing stage.

The decorative techniques described and depicted by Koch (1998 p. 127) are various forms of simple impressions, simple incisions, incrustation, and modelled application. Her descriptions and depictions of simple impressions with 'whipped cord' lack the characteristic impressions of fibre and instead resemble excisions (cf. Koch 1998 p. 541 on catalogue nr. 213.4; see also Fig. 6.10). The bases of the depressions which result from these techniques are described as having irregular microrelief with tool traces (such as rope) preserved (cf. Koch 1998 Fig. 98). Therefore, these techniques were applied to wet clay. There is no description of the microrelief of the incisions but an application on wet or leather-hard clay seems most likely given the position of surface treatment in the production process (see below). The position of incrustations is kept at the same place as in Funnel Beaker West vessels, following the arguments in Section 6.2. Lastly, the descriptions and drawings of decorations in the catalogue of Koch (1998) suggest these decorative techniques only appear in certain orders. Simple impressions and simple incisions can cut across each other and across application and excision, but applications and excisions do not cut across simple impressions and simple incisions. In addition, no vessels show combinations of excisions and simple incisions.

Koch (1998 p. 127) argues for two forms of surface treatment. The first form is burnishing, as evidenced by vessel surfaces with vertical and horizontal striations, compact microtopography, and glossy aspect. These traces may or may not cut across the traces of decoration, hinting, together with the striation, at an application to leather hard clay. The second form of surface treatment is smudging (ibid. p. 129). Evidence for this surface treatment are vessels with a dark grey exterior surface and oxidised margins and reduced cores on breaks (see also Section 6.2).

No further operations are mentioned in Koch (1998) apart from the firing process. Therefore, we may assume drying to leather-hard occurs between (or during) the application of decorative techniques and surface treatment. Drying to a dry consistency most likely occurred between surface treatment and firing of vessels.

Lastly, Koch (1998 p. 129) argues for open firing of Funnel Beaker North vessels, based mostly on inconsistencies in the archaeological evidence for kilns from this period and ethnographic evidence. This interpretation is followed here (see also Section 6.1).

The overview of ceramic technology in Koch (1998) exhibits medium to high fragmentation and qualitative data. For example, Koch (1998) describes the sequence of roughing-out techniques (see above), and her documentation enables a reconstructing of decorative techniques (see above), but operations during other stages of the ceramic production process are simply listed. There are no tallies with information about the frequency of most techniques. Therefore, these fragments and techniques are fully connected in the random seed for sim_FBN and the edge values have been set to binary (see Fig. 2; Tab. 1).

## 4 Random Seed for Sim_Odisha

The last random seed is for the outgroup sim_Odisha. This random seed is based on an ethnographic survey by Behura (1978) of ceramic production processes in the state of Odisha, India. The survey is part of a broader effort to document these technical traditions (cf. Mahias 1993; Saraswati and Behura 1966).

The nature of this ethnographic dataset is different from the archaeological datasets discussed above. Behura (1978) provides descriptions of the techniques, tools,

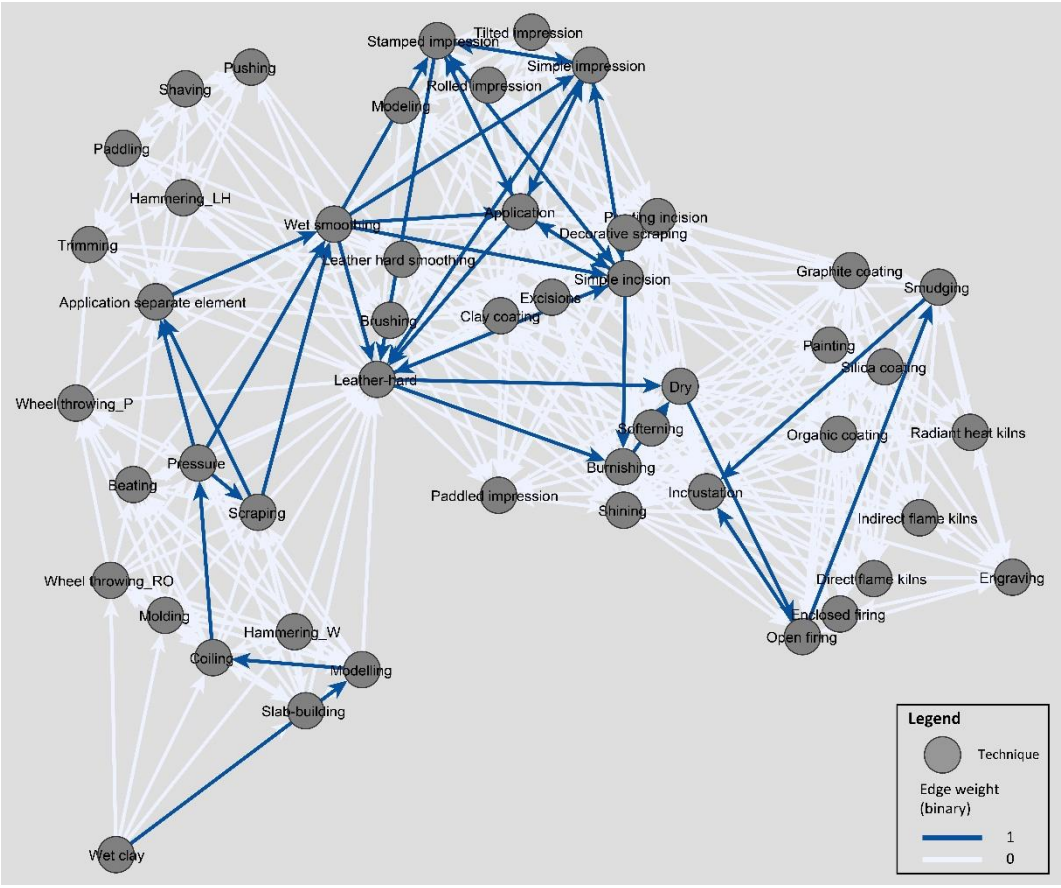



Figure 2: Random seed for body of knowledge sim_FBN. The blue edges indicate possible combinations of techniques based on the analyses of Funnel Beaker North vessels in bog depositions from Zealand, Møn, Lolland, and Falster (Denmark) by Koch (1998).

gestures, and postures involved in ceramic production, rather than interpretations of technical traces. This means less interpretation is required while translating the data into the vocabulary from Roux (2019). As such, the discussion below focuses on the connections between the various path segments and techniques discussed by Behura (1978). However, there are two exceptions for which more interpretation is required to arrive at the technical operations.

Firstly, there is no mention of finishing techniques in Behura (1978). However, the application of a wet cloth to vessels after preforming is frequently described (ibid. pp. 136; 153; 175). Therefore, the finishing techniques probably involve smoothing of wet or leather-hard clay with a soft tool after preforming and prior to surface treatment or decoration.

Secondly, Behura (1978 p. 168) refers to stamped impressions made on leather-hard clay. However, stamping leather hard clay is not possible in the framework of Roux (2019 p. 106). As such, the stamped impressions are translated here as paddled impressions. Paddled impressions can be made on leather-hard clay and as such are the closest mechanical equivalent to the actions described by Behura (1978).

Apart from these two exceptions, the interpretation of Behura's (1978) account of ceramic technology is comparatively straightforward.

Behura (1978 Ch. 5-6) describes several sequences of roughing-out, preforming, and finishing techniques. These sequences include roughing out and preforming vessels by means of wheel throwing, followed by wet smoothing, drying of the preform to a leather-hard consistency, and lastly a combination of paddling and/or shaving. Other sequences involve (combinations of) the roughing-out techniques modelling, moulding, and coiling, followed either by preforming through continuous pressure or beating, or by drying the clay to leather-hard consistency followed by paddling. Behura (1978 p. 155) specifically stresses the prominence of beating in the ceramic production processes in India. Lastly, handles or spouts may be applied separately to the vessel wall (cf. Behura 1978 p. 206). The evidence for the operations during the subsequent finishing stage has been discussed above.

Behura (1978) disconnects these sequences of techniques in the roughing-out and preforming stage from techniques during surface treatment and decoration. According to Behura (1978), there are four separate methods to apply decorations, which all occur while the vessel is leather-hard: simple incision (ibid. p. 168), paddled impression (see above), modelled application (ibid. p. 170), and a combination of excision, incrustation, and burnishing (ibid. p. 170). Some vessels are painted after firing (ibid. p. 208). The surface treatment techniques clay coating, burnishing and shining would be common for leather hard vessels (ibid. pp. 163, 166-7), whereas the application of silica coating (ibid. pp. 210-1) and organic coating (ibid. pp. 167-8) are rare, and always applied to fired vessels. Behura (1978) documents no further operations on dry clay.

Three types of firing architecture were in use among potters in Odisha according to Behura (1978 Ch. 8). These types of firing architecture are open firing, enclosed firing, and indirect flame kilns.

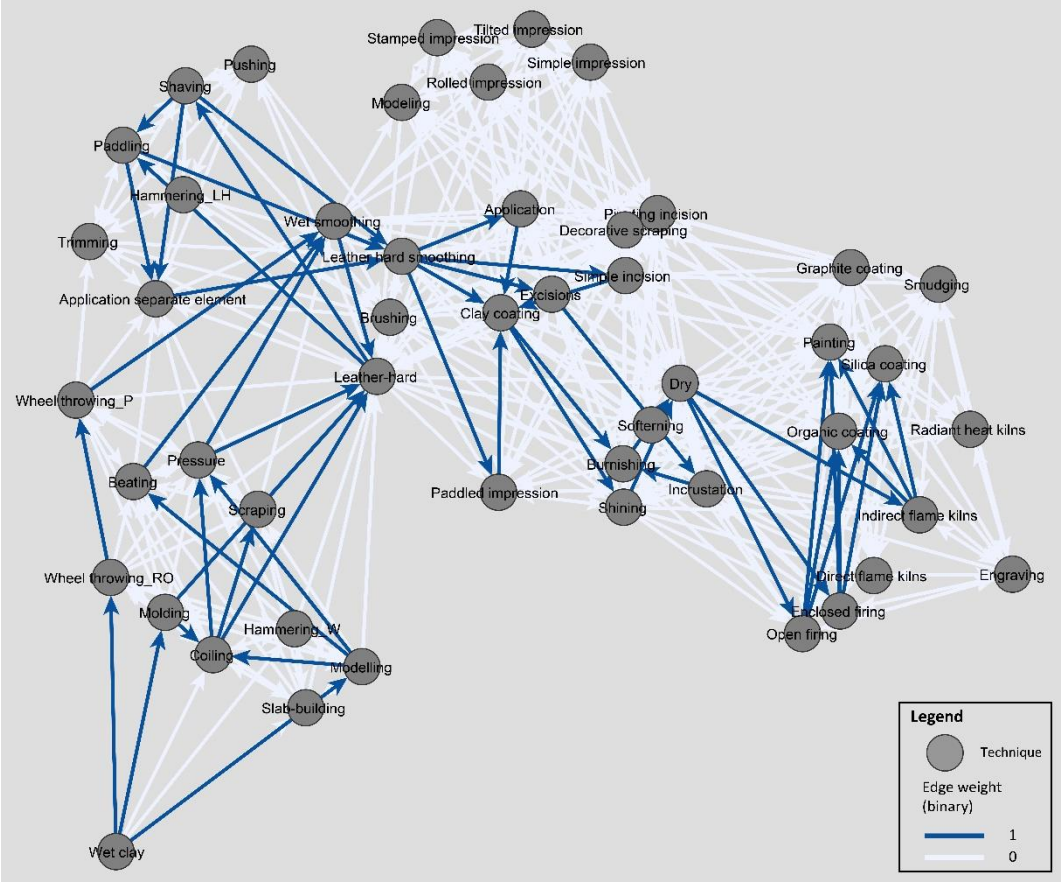



Figure 3: Random seed for sim_Odisha. The event space is based on Behura's (1978) survey of ceramic production techniques in Odisha, India. The value of the edges is binary, given the absence of quantitative data in the survey.

Figure 3 presents the random seed based on Behura's (1978) survey of ceramic production techniques (see above, Tab. 1). As shown above, the dataset has medium to high fragmentation with only qualitative descriptions. As such, the random seed has binary edge values.

## 5 Overview

This appendix is an overview of the simulation process and choices behind the sister groups sim_Vl and sim_FBN and the outgroup sim_Odisha in Chapter 10. The simulation process applies the algorithm for guided random path generation from Section 4.2 to three separate event spaces for ceramic production among Vlaardingen groups, Funnel Beaker North groups, and potters in modern Odisha (India). These event

spaces, or random seeds, are based on datasets about ceramic production from Stet (2021), Koch (1998), and Behura (1978), respectively. These datasets are translated into the vocabulary from Roux (2019) and plugged into the total *chaîne opératoire* as (fully connected) path segments with binary or absolute edge weights, depending on the fragmentation of the ceramic production process and the presence of quantitative data in the base dataset. As such, the random seeds are an overview of the potential variation in ceramic production processes based on fragmented and sparse data. The algorithm for guided random path generation then generates probable specific *chaînes opératoires* based on this overview of potter's knowledge.

The procedure applied above to datasets for Vlaardingen, Funnel Beaker North, and contemporary Indian ceramic production could be applied in many other contexts to overcome similar issues with sparse or fragmented data. As such, guided random path generation is a broadly applicable tool which enables probabilistic comparisons of legacy datasets or datasets which lack complete specific *chaînes opératoires* due to high fragmentation rates.

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